Problem 1.10

Using compound interest formula, how long would it take for an investment of \$15,000 to increase to \$45,000 if the annual compound interest rate is 2%?

Solution

Using the compound interest formula

$$A = P(1+i)^t,$$

substitute the given values:

$$45{,}000 = 15{,}000(1.02)^t.$$

Divide both sides by 15,000:

$$3 = (1.02)^t.$$

Taking natural logarithms on both sides,

$$\ln(3) = \ln\left((1.02)^t\right) = t\ln(1.02).$$

Therefore,

$$t = \frac{\ln(3)}{\ln(1.02)}.$$

Calculating,

$$t \approx \frac{1.098612}{0.019803} \approx 55.478.$$

Hence, the investment will take approximately

$$\boxed{55.478 \text{ years}}.$$